Computer Vision Based Parking Authorization & Authentication System

Final Year Project Thesis

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Project Overview

Many parking systems use RFID cards to open barriers. Although this system is quick and affordable, but an RFID only checks the card rather than the vehicle itself. The RFID card can be shared or stolen, allowing another driver to enter using the same credential. This project addresses that weakness by checking both the RFID card and the vehicle‘s number plate before allowing access to the parking area. It was developed for places such as universities and offices, with the aim of improving parking security while keeping equipment costs low.

System Design and Implementation

The hardware is built around an Arduino UNO microcontroller and two MFRC522 RFID readers: one at the entrance and one at the exit. When a card is scanned, the Arduino sends the card ID and gate direction to a Python application. Instead of using a separate camera for each lane, the project uses one camera on a servo driven bracket. The camera turns to face the entrance or exit, depending on which RFID reader is used. A small LCD displays messages such as "SCANNING," "ACCESS GRANTED" or "ACCESS DENIED." A 4x4 keypad also provides an emergency bypass for emergency situations, and its use is recorded for later review.

Hardware Components

Hardware Components	Purpose
Arduino microcontroller	Controls the hardware and communicates with Python
Two MFRC522 RFID readers	Read credentials at entry and exit
RFID tags	Identify registered users
Camera	Captures vehicle license plate images
Camera positioning servo	Adjusts the camera direction
Barrier servo motor	Controls the access barrier
LCD with I2C interface	Displays system status and messages
4x4 keypad	Allows emergency bypass entry
Laptop	Runs the Python application and database
USB connection	Transfers data between Arduino and computer

Image Processing and Access Checks

The Python application uses OpenCV and EasyOCR to read the vehicle‘s number plate. Because the camera moves between lanes, the software waits for it to settle and clears old video frames that may be blurred. It then zooms in on the plate, converts the image to grayscale, reduces noise with a bilateral filter, and uses adaptive thresholding to make the characters clear. EasyOCR detects the text and recognises the letters and numbers. The software then removes spaces, punctuation and common regional words, such as "SINDH" or "PUNJAB," before checking the result against the registered plate number.

Python Libraries and Modules

Library / Module	Purpose
OpenCV	Camera capture and image processing
EasyOCR	Extracting text from vehicle license plates
NumPy	Numerical operations and image array handling
PySerial	Communication between Python and Arduino
SQLite3	Storing authorised users and access records
re	Cleaning and validating recognised text
os	Operating system and file handling operations
ssl	Security related functions for encrypted connections.
time	Time related operations and performance measurement
datetime	Creating timestamps for access records

Access Control and Database

A local SQLite database stores each authorised user‘s RFID ID, number plate and current inside/outside status. It also keeps a time-stamped record of access attempts, including successful entries, mismatched plates, passback denials and emergency overrides. This status check is central to the system‘s anti-passback feature. For example, if a user has already entered the parking area, the same RFID card cannot be used to authorise another entry before the system records an exit. The Python software communicates with the Arduino over a serial connection and uses asynchronous processing so that waiting for camera movement does not stop other system tasks.

Testing and Results

The thesis reports tests of the camera, serial communication, number-plate reading and database logic. In a test of 50 plate images, EasyOCR achieved a 62% exact-match rate on raw images. After the OpenCV processing steps were applied, the rate increased to 94.5%. The text-detection stage located plate text correctly in 99% of cases, while the report‘s results table gives a 99.6% success rate for removing regional text and other unwanted characters. The remaining recognition errors were mainly linked to heavy obstruction or unusual plate fonts.

Timing and Reliability

The average serial communication delay was under 2.5 milliseconds. The camera needed about 1.2 seconds to settle, followed by an average of 450 milliseconds for image processing and the database check. The reported total time from card scan to gate response was around 1.65 seconds. In a stress test of 50 consecutive authentications, the database recorded all events without dropped transactions or data collisions, and the anti-passback tests successfully rejected repeated entry attempts that broke the required entry/exit order.

Conclusion and Future Work

The project demonstrates how RFID, number-plate recognition and local database checks can work together to make parking access more secure. Its main practical contribution is using one movable camera for two lanes, while keeping access decisions and records on the local system. The results show that image preparation greatly improves recognition, although difficult lighting, blocked characters and non-standard fonts remain challenges. The thesis recommends future work on a shared database for multiple gates, traffic-pattern analysis using past access logs, moving the software to a compact edge computer, and adding vehicle detection so OCR runs only when a vehicle is present. Overall, the project provides a working foundation for a lower-cost and more accountable automated parking system.